

Reconstructing the Experience of Nüshu Culture: An Exploration via Multimodal Mixed Reality Systems

Zheyu Feng
Communication University of China
Beijing, China
fengzheyu2002@cuc.edu.cn

Boya Liu
Savannah College of Art and Design
Savannah, USA
boyliu22@student.scad.edu

Zhonghe Ruan
Communication University of China
Beijing, China
rzh0405@cuc.edu.cn

Xinyi Zhang*
Communication University of China
Beijing, China
xzhang1206@cuc.edu.cn

Zihan Gao*
Communication University of China
Beijing, China
zihan@cuc.edu.cn

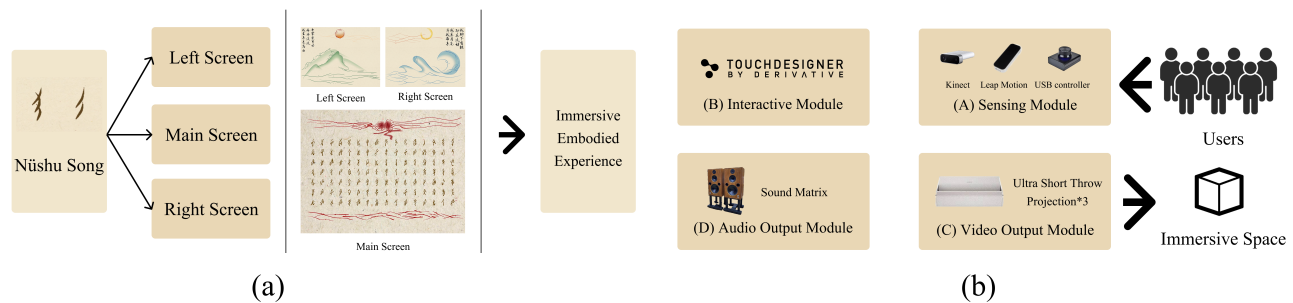


Figure 1: (a) The construction process of multimodal interaction scenarios. (b) The technical framework for the immersive system.

Abstract

Nüshu (Jiangyong Nüshu), a script developed and practiced exclusively by women in China, holds recognition as National Intangible Cultural Heritage. This paper presents the design and implementation of an immersive multimodal interaction system centered on *The Song of Nüshu*, a foundational cultural artifact. By integrating Natural Interaction (NI) and Mixed Reality (MR) technologies, the system constructs a visual environment inspired by classical Chinese landscape painting aesthetics. Within this experiential space, users engage with representations of pivotal female historical figures and events interwoven with Nüshu textual elements. This framework enables participant-driven narrative construction that foregrounds Nüshu's aesthetic dimensions and socio-cultural significance. Our work advances a heritage preservation methodology that innovatively reinterprets tradition while explicitly foregrounding women's historical agency and expressive practices.

*Xinyi Zhang and Zihan Gao are corresponding authors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

MM '25, Dublin, Ireland

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2035-2/2025/10

<https://doi.org/10.1145/3746027.3756145>

CCS Concepts

• Applied computing → Media arts; • Human-centered computing → User centered design.

Keywords

Nüshu, Intangible Cultural Heritage, Multimodal Interaction, Mixed Reality

ACM Reference Format:

Zheyu Feng, Boya Liu, Zhonghe Ruan, Xinyi Zhang, and Zihan Gao. 2025. Reconstructing the Experience of Nüshu Culture: An Exploration via Multimodal Mixed Reality Systems. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3756145>

1 Introduction

Nüshu, an aesthetically distinctive script, embodies the creative resilience of Chinese women within historical patriarchal constraints. Beyond its linguistic function, it served as a vital medium for emotional expression and cultural preservation. Its unique form and complex historical trajectory endow it with profound cultural significance, despite enduring cycles of marginalization.

Contemporary efforts to revitalize Nüshu as living heritage demand urgent scholarly intervention [2]. Emerging media technologies, particularly computational graphics and interactive systems, offer transformative frameworks for recontextualizing traditional

culture. Prior research demonstrates that such technologies reconfigure cultural engagement, shifting audiences from passive consumption to participatory co-creation, thereby amplifying both interactivity and cultural resonance. For instance, Draper et al. engineered a multi-user mixed reality system enabling collaborative exploration of digitally reconstructed heritage sites [1]. Similarly, Xu et al. established that multimodal mixed-reality environments enhance cultural learning through multisensory interaction, fostering deeper cognitive and affective connections to heritage narratives [3].

Aligned with initiatives to reinterpret Nüshu for modern audiences [4], this paper proposes an immersive, multimodal interactive space integrating natural interaction paradigms with mixed reality. By synthesizing tradition with emergent media, this framework aims to catalyze cultural participation among youth, revitalizing awareness, identity, and continuity of this intangible heritage.

2 User Experience Design

The overall system consists of three surrounding projection screens and a central interactive area, as shown in Figure 1. Through multimodal interaction, the installation presents both the content and cultural significance of The Song of Nüshu.

2.1 Scenario Design

The main screen serves as the system's visual core, initially presenting a 7×16 grid of Nüshu characters excerpted from The Song of Nüshu. This text captures how women in secluded communities historically used writing to express emotions and document life, with themes such as marriage, friendship, and fate reflecting shared female experiences under patriarchal constraints, as shown in Figure 2. The two side screens expand this narrative through landscape compositions formed by Nüshu characters. These poetic visuals metaphorically depict women as mountains, oceans, and celestial bodies, symbolizing female strength and their connection to nature. Users are immersed in a culturally rich soundscape upon entering the virtual space. A Nüshu song in the local dialect, performed by Zhou Jing'er — a cultural inheritor — blends traditional chant with regional tone, deepening emotional and cultural engagement. As interaction progresses, such as rotating a virtual turntable, Nüshu characters fall like sand, while modern female vocals replace the traditional song. This shift forms a trans-temporal cultural narrative, connecting historical expression with contemporary reinterpretation. The experience serves as a multisensory tool for preserving, reactivating, and reimagining intangible cultural heritage.

2.2 Interaction Design

When users approach the main screen, Nüshu characters within their presence gradually light up and are translated into modern Chinese. This interaction — “to be seen is to be awakened” — symbolizes cultural reactivation through reinterpretation. A second interaction involves rotating a physical slider, causing the characters to disintegrate like sand — a metaphor for Nüshu's historical decline and the loosening of constraints on female expression. As the characters disappear, a final animation titled “Female Empowerment” is triggered, emphasizing the evolving spirit of Nüshu and

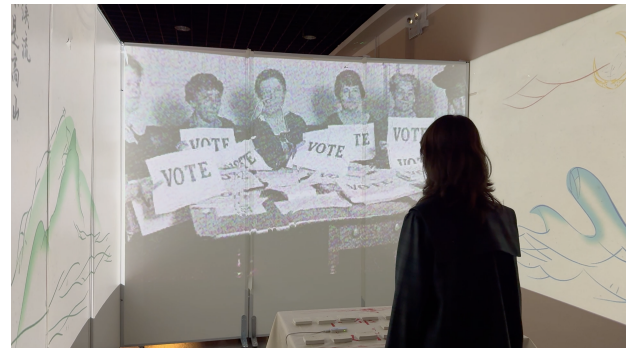


Figure 2: On-site exhibition.

its relevance to contemporary narratives of gender and cultural identity.

3 Technical Implementation

The visual and interactive components of this system were developed using Touch Designer. As illustrated in Figure 1, the system architecture follows a modular workflow. Upon entering the interactive area, (a) the sensor module collects data on the number of participants, their positions, and gestures using devices such as Kinect, Leap Motion, and the iWit USB Controller. This data is then transmitted to (b) the interactive content control system, which processes the input and generates corresponding real-time visual and behavioral responses. These responses are subsequently sent to both (c) the audio output module and (d) the projection output module, which together render and deliver the digital content within the exhibition space. This integrated system ensures responsive, real-time interaction between the audience and the immersive environment.

4 Conclusion

Multimodal mixed reality systems hold application potential in the inheritance and reinterpretation of Nüshu culture. By integrating text, sound, and interaction, these systems create immersive and participatory cultural spaces where traditional female expressions are revived within contemporary forms of expression. The combination of visual and auditory metaphors with user behavior stimulates emotional resonance and cultural reflection. Augmented, virtual, and mixed reality technologies have become pivotal in cultural heritage preservation, offering novel means to visualize, interpret, and engage with intangible heritage while overcoming physical and temporal barriers. This approach not only preserves intangible cultural heritage but also promotes its continuous evolution, allowing marginalized voices to reemerge in contemporary contexts and highlighting the proactive role of digital technologies in cultural transmission.

Acknowledgments

This project is funded by National Key Research and Development Program of China (No. 2022YFF0902305) and Fundamental Research Fund for the Central Universities in China (No. CUC25SG005).

References

- [1] Heather Draper, Bertolini Giacomo, Bekiari Kalliopi, and Papaioannou Georgios. 2024. Reimagining Historical Exploration: Multi-User Mixed Reality Systems for Cultural Heritage Sites. *Applied Sciences* 15, 5 (2024), 2854. doi:10.3390/app15052854
- [2] Congrong Li. 2024. History, Characteristics, and Modern Vitality of Nüshu: A Cultural Anthropology Perspective. In *Anthropological Explorations of Gender, Identity, and Economics*. 85.
- [3] Yifan Xu, Hongyu He, and Wei Zhang. 2021. The Influence of Collaborative and Multi-Modal Mixed Reality: Cultural Learning in Virtual Heritage. *Multimodal Technologies and Interaction* 5, 12 (2021), 79. doi:10.3390/mti5120079
- [4] Ge Yu, Shamim Akhter, Tribhuwan Kumar, Geovanny Genaro Reivan Ortiz, and Kundharu Saddhono. 2022. Innovative application of new media in visual communication design and resistance to innovation. *Frontiers in Psychology* 13 (2022), 940899.